Multiple Choice: Write the **CAPITAL LETTER** corresponding to the correct answer on the line provided.
[1 Mark Each – 5 Marks Total]

1. If $\tan(\theta)=\frac{2}{5}$ and $\theta\in\left[0,\frac{\pi}{2}\right]$, then the **exact** value of $\cos(2\theta)$ is

-1 X A

A. $\frac{5}{\sqrt{29}}$ B. $\frac{21}{29}$ C. $\frac{10}{\sqrt{29}}$ D. $-\frac{4}{29}$

2. The expression $\sin\left(\frac{\theta}{5}\right)\cos\left(\frac{2\theta}{7}\right)-\sin\left(\frac{2\theta}{7}\right)\cos\left(\frac{\theta}{5}\right)$ is equivalent to

D

A. $\cos\left(\frac{17\theta}{35}\right)$ B. $\sin\left(\frac{17\theta}{35}\right)$ C. $\sin\left(\frac{3\theta}{35}\right)$ D. $-\sin\left(\frac{3\theta}{35}\right)$

3. Which of the following is false?

C

A. One radian is approximately equal to $57.3°$.

B. $\csc(\theta)$ is always negative where $\frac{3\pi}{2}<\theta<2\pi$.

C. A co-terminal angle to $\frac{7\pi}{12}$ is $\frac{115\pi}{12}$.

D. $\cot\left(\frac{5\pi}{4}\right)$ is 1.

4. When $\cos\left(\frac{4\pi}{3}\right)$ is written as a function of its co-function angle the result is

C

A. $\sin\left(\frac{3\pi}{2}\right)$ B. $\cos\left(\frac{3\pi}{2}\right)$ C. $-\sin\left(\frac{\pi}{6}\right)$ D. $\sin\left(\frac{\pi}{6}\right)$

5. Given $\sin(A)=\frac{7}{8}$ and $\cos(B)=\frac{4}{5}$, where A and B are acute angles, then the exact value of $\cos(A-B)$ is equal to

-1 X D

A. $\frac{3}{8}$ B. $\frac{16}{25}+\frac{49}{64}$ C. $\frac{4\sqrt{15}+21}{40}$ D. $\frac{28-3\sqrt{15}}{40}$

6. If $\tan(\theta)=-\frac{4}{5}$, where $\theta\in\left(\frac{\pi}{2},\pi\right)$, determine the **exact** value of $\sin(2\theta)$. **[3 Marks]**

-3

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7. Determine the **exact** value of the following. **[6 Marks]**

a. $\sin\left(\frac{19\pi}{12}\right)$ **[3 Marks]**

-3

b. $\cos\left(\frac{13\pi}{8}\right)$ **[3 Marks]**

$\cos\left(\frac{13\pi}{8}\right)$

$=\cos\left(\frac{3\pi}{2}+\frac{13\pi}{8}\right)$

$=\sin\left(\frac{13\pi}{8}\right)$

-2.5

8. Rewrite each of the following as its equivalent **co-function identity** and then **evaluate**. **[4 Marks]**

a. $\cos\left(\frac{5\pi}{6}\right)$ **[2]**

$\cos\left(\frac{5\pi}{6}\right)=-\sin\left(\frac{\pi}{2}+\frac{5\pi}{6}\right)$

$=-\sin\left(\frac{3\pi}{6}+\frac{5\pi}{6}\right)$

$=-\sin\left(\frac{8\pi}{6}\right)$

$=-\sin\left(\frac{4\pi}{3}\right)$ → break further

$=-\frac{1}{2}$

-1

b. $\csc\left(\frac{11\pi}{4}\right)$ **[2]**

$\csc\left(\frac{11\pi}{4}\right)=\sec\left(\frac{\pi}{2}+\frac{11\pi}{4}\right)$

$=\sec\left(\frac{2\pi}{4}+\frac{11\pi}{4}\right)$

$=\sec\left(\frac{13\pi}{4}\right)$

-1

THINKING – [5 MARKS]

1. Prove: $\dfrac{\sin(3x)-\sin(x)}{\cos^2(x)-\sin^2(x)}=2\sin(x)$. **[5 Marks]**

-5

*** 2 Marks are awarded in the Communication Category for the use of correct mathematical form. ***

1. **Completely** simplify $\dfrac{\sin^2\left(\frac{3\pi}{2}-x\right)-\tan\left(\frac{3\pi}{4}\right)\cos^2\left(\frac{\pi}{2}+x\right)}{\sin^2(\pi-x)+\cos(\pi-x)\sin\left(\frac{\pi}{2}+x\right)}$. **[4 Marks]**

$$\frac{\sin^2(\frac{3\pi}{2}-x)-\tan(\frac{3\pi}{4})\cos^2(\frac{\pi}{2}+x)}{\sin^2(\pi-x)+\cos(\pi-x)\sin(\frac{\pi}{2}+x)}$$

$$=\frac{\cos^2(x)-\tan(\frac{3\pi}{4})[-\sin^2(x)]}{\sin^2(\pi)\cos^2(x)-\cos^2(\pi)\sin^2(x)+\cos(\pi)\cos(x)+\sin(\pi)\sin(x)}$$

$$=\frac{\cos^2(x)+\sqrt{3}\,[-\sin(x]}{(0)\cos^2(x)-(-1)\sin^2(x)+(-1)\cos(x)+(0)\sin(x)}$$

$$=\frac{\cos^2(x)+\sqrt{3}\,[-\sin(x)]}{\sin^2(x)-\cos(x)}$$ (rest of work on scrap paper)

-2.5

please correct

2. Determine the **exact** value of $\sin\left(\frac{29\pi}{72}\right)\sin\left(\frac{13\pi}{72}\right)-\cos\left(\frac{29\pi}{72}\right)\cos\left(\frac{13\pi}{72}\right)$. **[4 Marks]**

$\sin(\frac{29\pi}{72})\sin(\frac{13\pi}{72})-\cos(\frac{29\pi}{72})\cos(\frac{13\pi}{72})$

$=-\cos(\frac{29\pi}{72})\cos(\frac{13\pi}{72})+\sin(\frac{29\pi}{72})\sin(\frac{13\pi}{72})$

$=-[\cos(\frac{29\pi}{72})\cos(\frac{13\pi}{72})-\sin(\frac{29\pi}{72})\sin(\frac{13\pi}{72})]$

$=-[\cos(\frac{29\pi}{72}+\frac{13\pi}{72})]$

$=-[\cos(\frac{42\pi}{72})]$

$=-[\cos(\frac{7\pi}{12})]$

$=-\cos(\frac{7\pi}{12})$

$=-\cos(\frac{\pi}{3}+\frac{\pi}{4})$

-2

should have continued.

3. Prove the identity $(\sin^4 B+\cos^4 B)(\tan B+\cot B)^2=\frac{\sin^2 B}{\cos^2 B}+\frac{\cos^2 B}{\sin^2 B}$. **[4 Marks]**

$(\sin^4 B+\cos^4 B)(\tan B+\cot B)^2=\frac{\sin^2 B}{\cos^2 B}+\frac{\cos^2 B}{\sin^2 B}$

L.S. $=(\sin^4+\cos^4 B)(\tan B+\cot B)^2$

$=(\sin^4 B+\cos^4 B)\left(\frac{\sin B}{\cos B}+\frac{\cos B}{\sin B}\right)^2$

$=(\sin^4 B+\cos^4 B)\left(\frac{\sin^2 B}{\sin B\cos B}+\frac{\cos^2 B}{\sin B\cos B}\right)^2$

$=(\sin^4 B+\cos^4 B)\left(\frac{\sin^2 B+\cos^2 B}{\sin B\cos B}\right)^2$

(rest of work on scrap paper)

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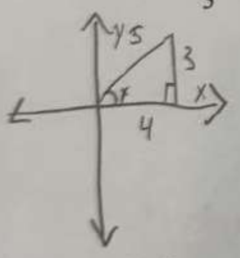
4. If $\sin x - \sqrt{3}\cos x = S\cos(x+T); S > 0$ and $0 < T < 2\pi$, then **find S & T**. **[5 Marks]**

−5

THINKING – [5 MARKS]

1. If $\cot(x) = \frac{4}{3}$, where $0 < x < \frac{\pi}{2}$, determine the **exact** value of $\cos(4x)$. **[5 Marks]**



$\tan(x) = \frac{3}{4}$

$a^2 + b^2 = c^2$

$4^2 + 3^2 = c^2$

$16 + 9 = c^2$

$25 = c^2$

$\sqrt{25} = c$

$c = 5$

$\cos(x) = \frac{4}{5}$

$\cos(4x) = \frac{16}{5}$

−3.5

$$(y)+\sqrt{3}\left[\cancel{-\sin^2(x)}\right]$$

$$\frac{}{\cancel{\sin^2(x)} - \cancel{\cos(x)}}$$

$$\rightarrow \frac{\cos^2(x)+\sqrt{3}\left[\cancel{-\sin^2(x)}\right]}{\sin^2(x)-\cos x}$$

$$=-\cos(x)-\sqrt{3}$$

$$=(\sin^4 B+\cos^4 B)\left(\frac{1}{\sin B\cos B}\right)^2 ✓$$

$$=(\sin^4 B+\cos^4 B)\left(\frac{1^2}{\sin^2 B\cos^2 B}\right) ✓$$

$$=(\sin^4 B+\cos^4 B)\left(\frac{1}{\sin^2 B\cos^2 B}\right)$$

$$=\frac{\sin^4 B+\cos^4 B}{\sin^2 B\cos^2 B}$$

$$R.S.=\frac{\sin^2 B}{\cos^2 B}+\frac{\cos^2 B}{\sin^2 B}$$

$$=\frac{\sin^4 B}{\sin^2 B\cos^2 B}+\frac{\cos^4 B}{\sin^2 B\cos^2 B}$$

$$=\frac{\sin^4 B+\cos^4 B}{\sin^2 B\cos^2 B}$$

$L.S.=R.S.$

$\therefore$ The identity is true